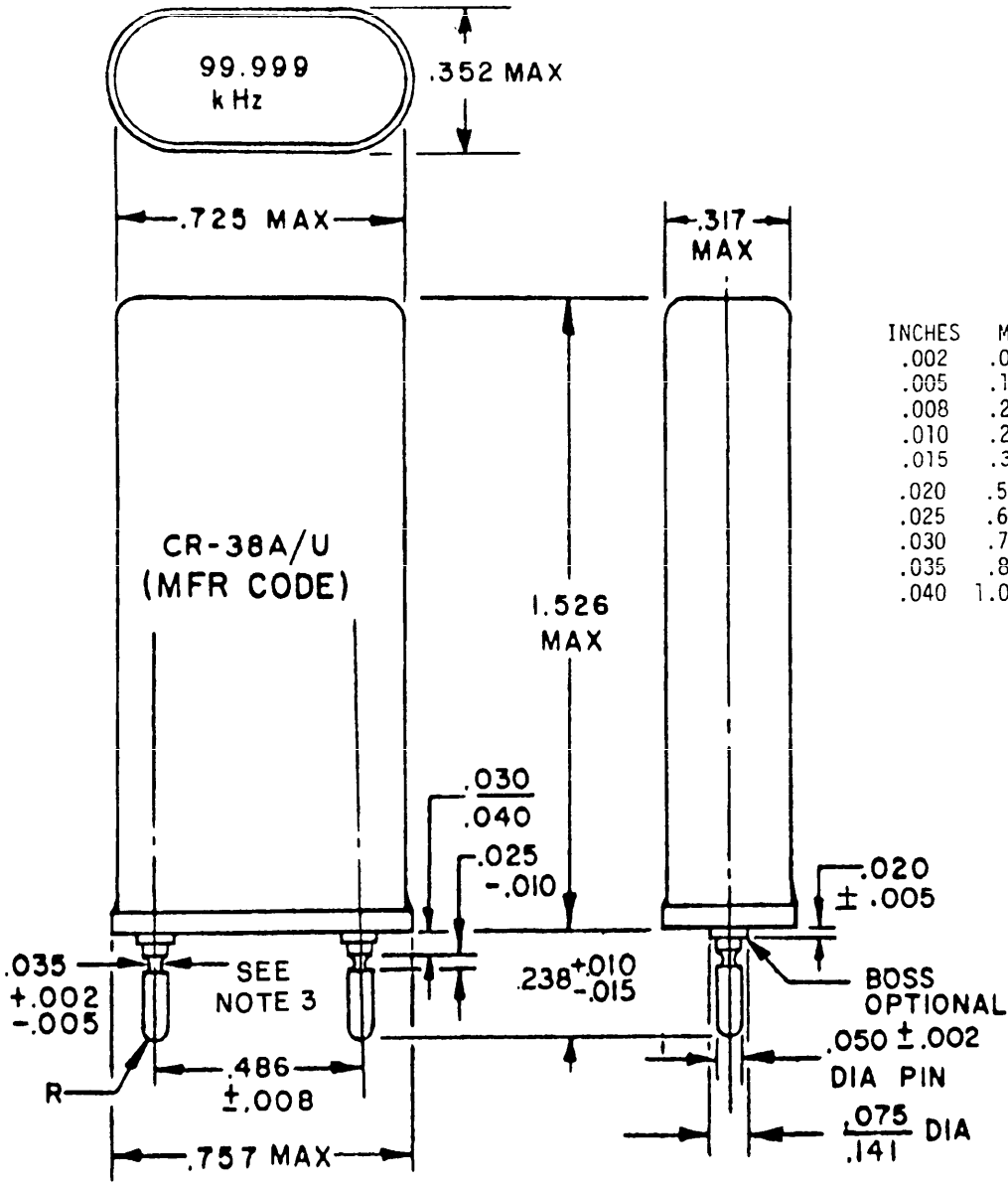
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-38A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the issue of MIL-C-3098.

Pertinent characteristics: 16 to 100 kHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.90
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	1.526	38.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Type CR-38A/U, crystal unit, quartz.

(D) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:21Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-13/U (MIL-H-10056/4)
Dimensions, marking, and configuration	See figure
Frequency range	16 to 100 kHz, inclusive
Capacitance, shunt	See table III
Frequency tolerance (operating temperature range)	±120 PPM
Equivalent resistance:	kHz — Ohms, maximum 16 to 50 — 110,000 50+ to 80 — 100,000 80+ to 100 — 90,000
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Antiresonance, load capacitance	20.0 ±0.5 pF
Operating temperature range	-40° to +70°C, inclusive
Shock (specified pulse):	
Frequency change permitted	±10 PPM
Equivalent-resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
(D) Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz — Grams, minimum 16 to 60 — 800 60+ to 100 — 700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volts	mW
TS-710/TSM	16 to 50	77,000	---	2.77	0.1
	50+ to 80	70,000	---	2.65	0.1
	80+ to 100	63,000	---	2.51	0.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:21Z
Check the source to verify that this is the current version before use.

TABLE III. Capacitance, shunt.

Frequency range, inclusive	Capacitance 1/
kHz	pF
16 to 34	$\frac{24}{\sqrt{f}} + 1.6$
34+ to 54	$\frac{33}{\sqrt{f}} + 1.6$
54+ to 100	$\frac{24}{\sqrt{f}} + 1.6$

1/ The letter "f" represents specified frequency in kHz. A capacitance tolerance of ±15% is permitted.

NOTES:
1. The socket designed to mate with crystal holder HC-13/U is Part No. M12883/04-01 (see MIL-S-12883/4).
   For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO guidance type NXT47.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SG, SM
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0606-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:21Z
Check the source to verify that this is the current version before use.